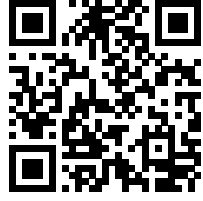


Contact & Collaboration:

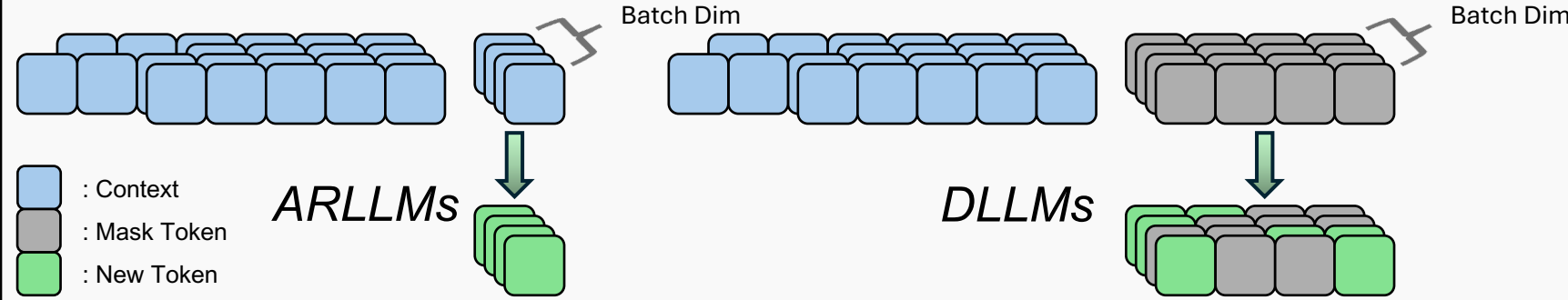
kaihua.liang@kaust.edu.sa

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marco@kaust.edu.sa



Background & Motivation

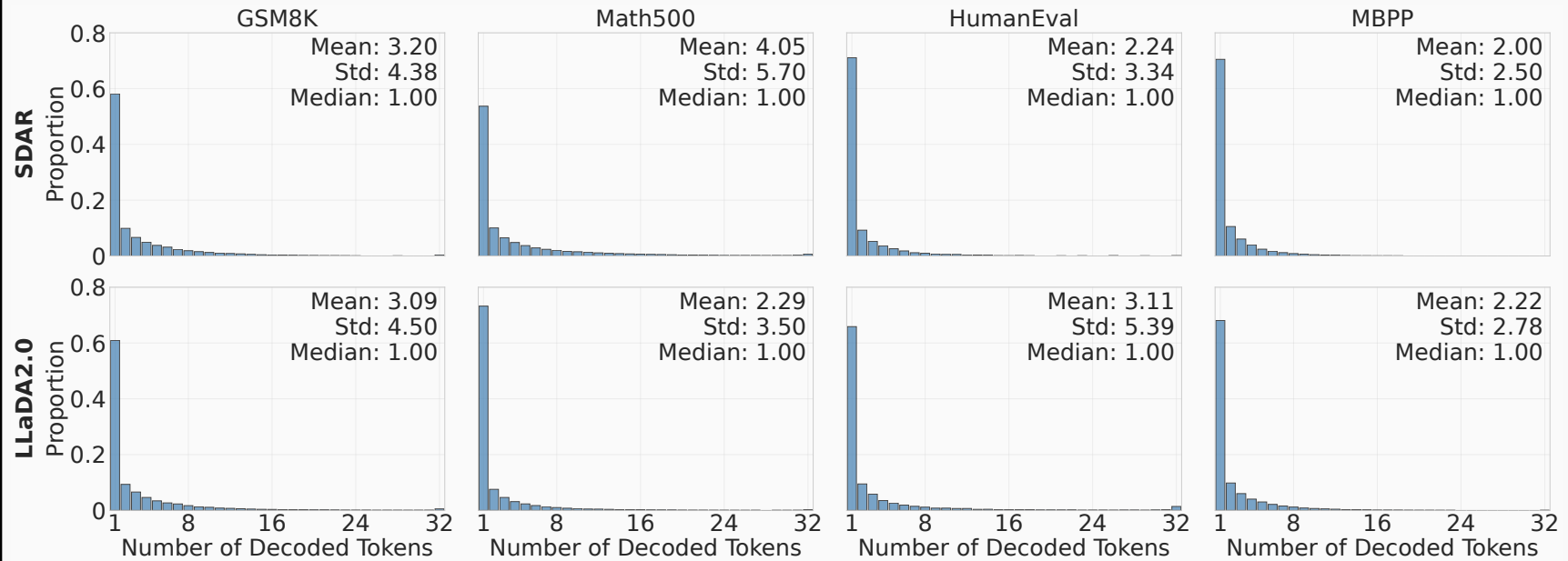


Diffusion LLMs offer a promising alternative paradigm to autoregressive LLMs. Yet, they represent a **computationally intensive workload**, even with block diffusion paradigm. Thus, their potential is limited by their **inference efficiency**.

FLOPs Formula forward of a layer of LLM:

$$Q \cdot (8h^2 + 4d_{ff}h + 4hL)$$

Q : Queries Length, h : Hidden Dimension, d_{ff} : Up-Projection Width, L : Context Length

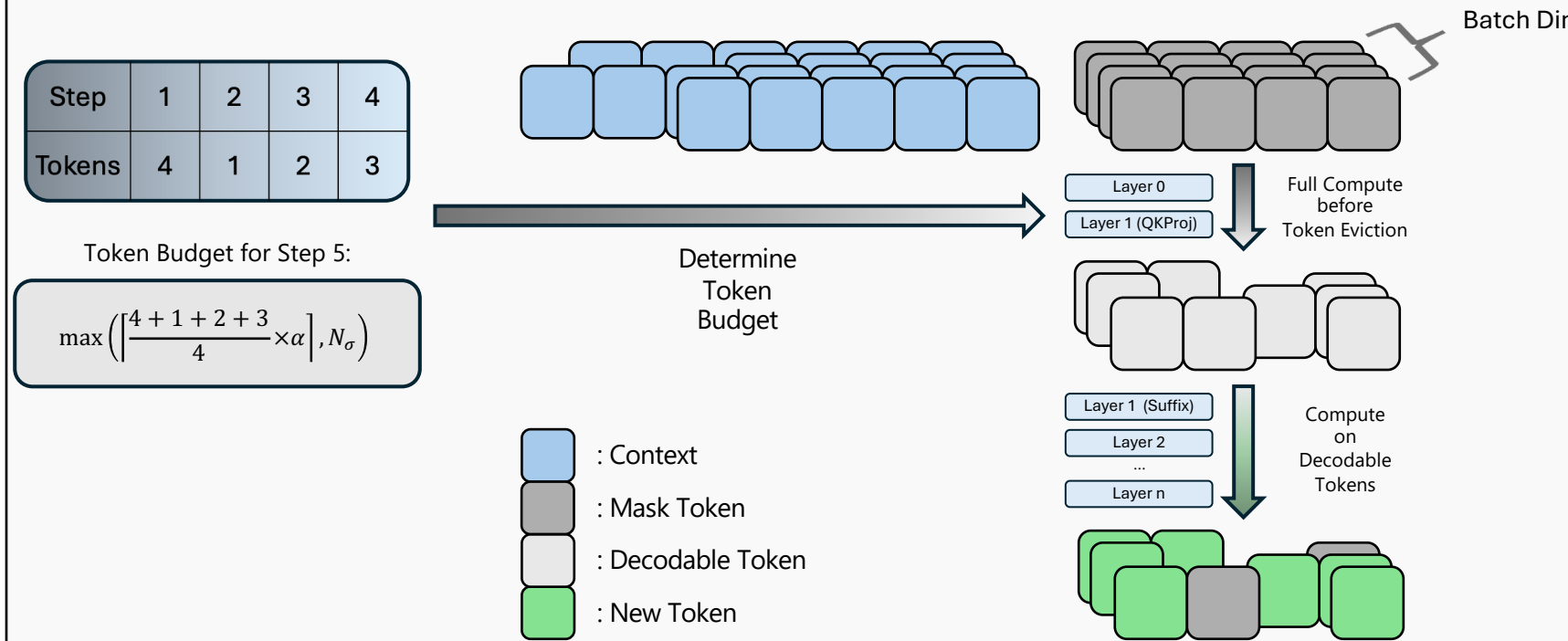


Decoding only a small portion per step -> **severe redundant computation**.

System Design

Comparison of DLLM inference paradigms. FOCUS further enhances the inference efficiency by predicting decodable tokens to eliminate computation redundancy.

METHOD	PARALLEL DECODING	KV CACHE	DECODABLE PREDICTION	INFERENCE EFFICIENCY
LLADA	✗	N/A	✗	★
FAST-dLLM	✓	APPROX.	✗	★★
SDAR / LLADA2.0	✓	EXACT	✗	★★★
FOCUS (Ours)	✓	EXACT	✓	★★★★



We propose the **FOCUS** system, being implemented on top of LMDeploy with customized Triton kernels, which consists of two key components:

- Importance-Based Token Eviction:** Preserve a dynamic budget of tokens with the **highest importance delta** in the first two layers.
- Delayed Cache with Neighbor Awareness:** Inside processing block, a token's KV are cache in the forward after both itself and its left neighbor are decoded.

Key Insight

Question: Is there a way to focus computation on decodable tokens?

Token Importance Delta

$\propto$

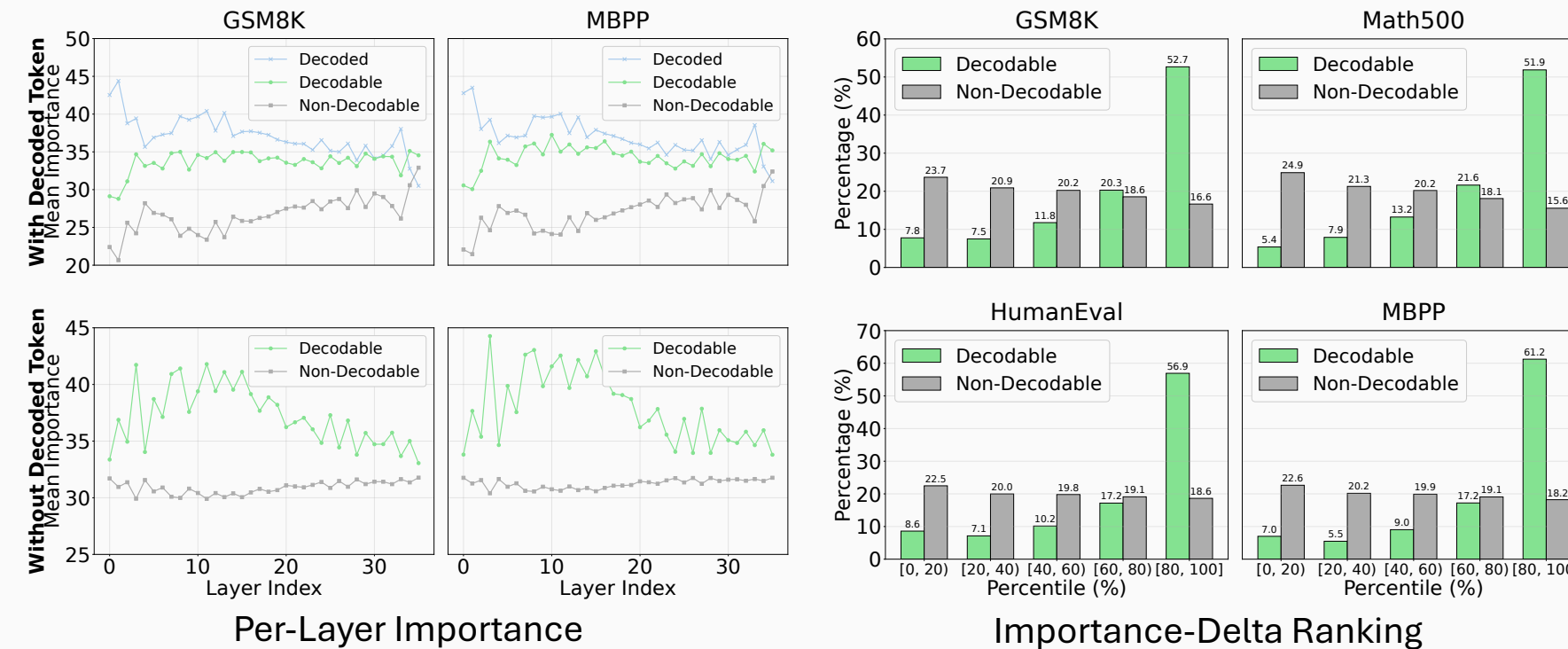
Decoding Probability:

$$\Delta I_j = \Delta_{L1-L0} \sum_{i,h} \text{Softmax}\left(\text{MaxPool1D}\left(S_{ij}^{(h)}\right)\right)$$

, where $S = \frac{(q_i^{(h)})^T k_j^{(h)}}{\sqrt{d_k}}$, $i, j \in \{1, \dots, B\}$ index the the intra-block B tokens.

Token	1	2	3	4	...	n
1	3.45	3.60	-2.12	-2.14	...	-2.80
2	2.04	2.06	0.77	-2.06	...	-2.81
3	0.73	1.75	0.92	-0.42	...	-2.95
4	-1.56	-0.36	2.38	2.11	...	-2.56
...	...	...	...	...	...	...
n	-0.93	2.35	2.35	2.62	...	6.52
	-0.40	6.07	2.21	4.04	...	-7.49

Column-Wise
Sums of
Delta Matrix

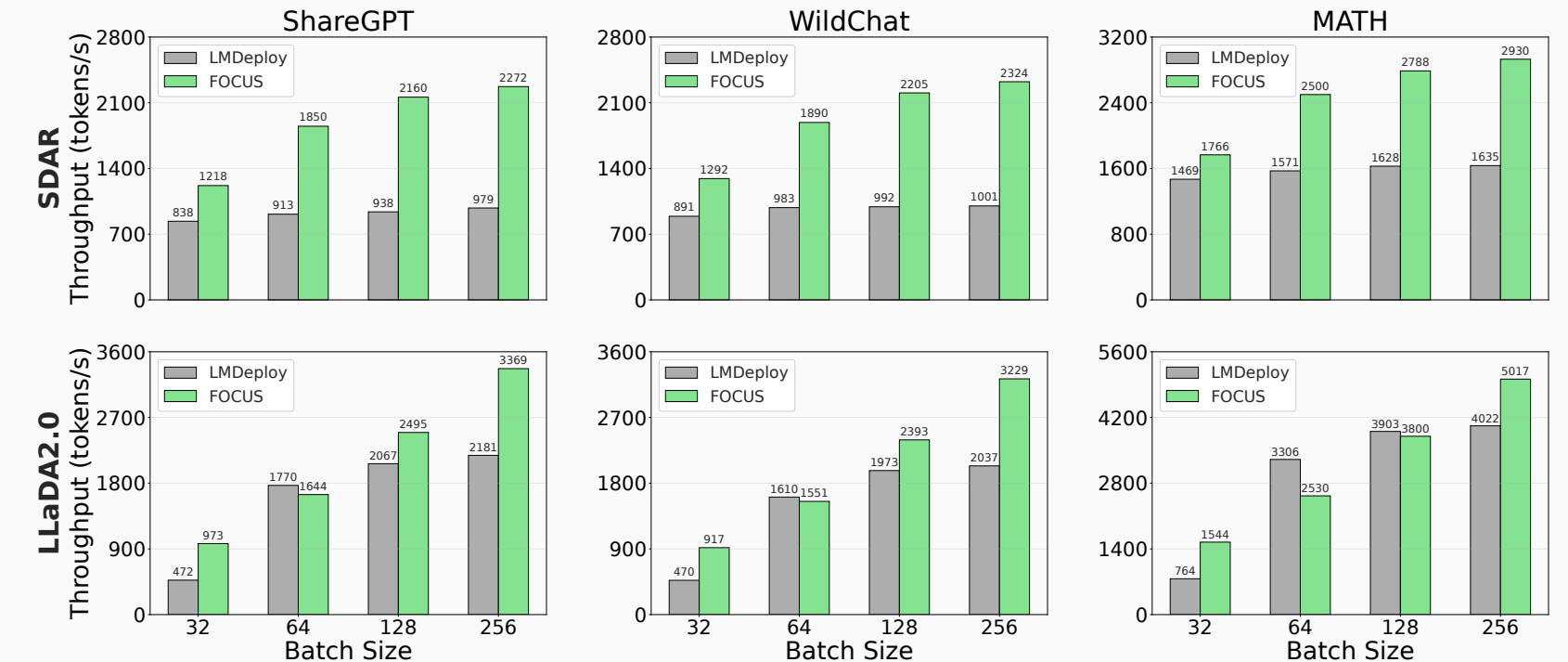


Evaluation

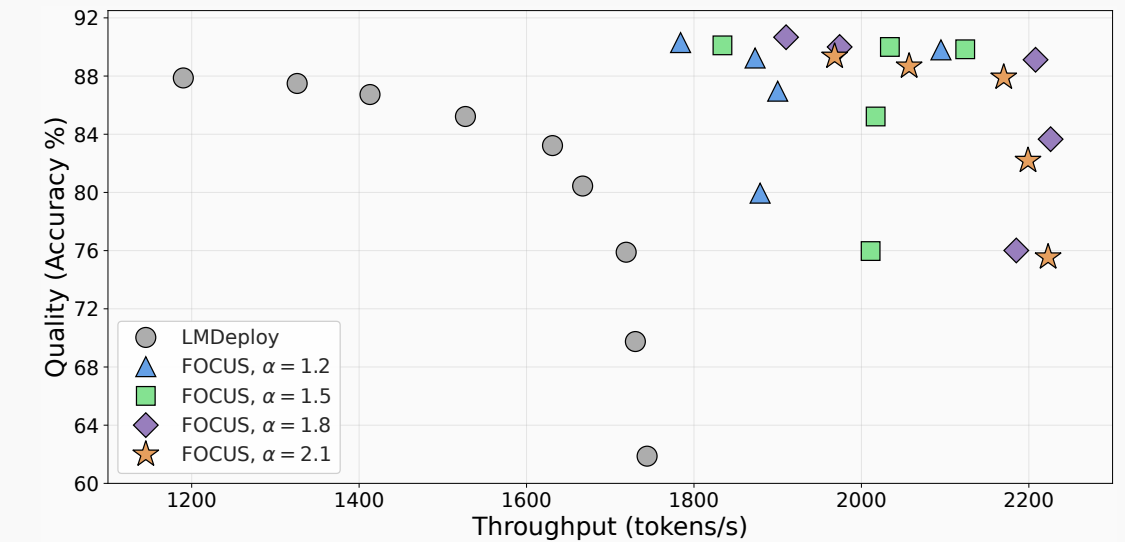
Improved generation quality under same confidence threshold.

Conf.	M	α	GSM8K	Math500	HumanEval	MBPP	IFEval
0.9	B	-	89.20	64.70	69.82	56.81	57.45
	F	1.2	89.84	64.60	72.56	58.17	60.97
	F	1.5	90.15	64.30	69.51	61.09	60.87
	F	1.8	90.75	63.80	71.34	58.95	60.11
0.8	B	-	87.57	59.30	65.85	50.78	58.69
	F	1.2	90.33	64.10	67.68	58.56	59.91
	F	1.5	89.73	62.20	69.21	59.73	60.97
	F	1.8	89.39	62.10	69.82	56.81	61.14
0.7	B	-	84.65	54.60	59.76	50.58	58.47
	F	1.2	89.24	62.00	68.29	56.81	61.91
	F	1.5	89.31	62.20	64.33	55.25	62.26
	F	1.8	88.03	59.80	64.33	53.89	60.08

Improved end-to-end throughput.



Improved quality-throughput tradeoff frontier.



Conclusion

We propose FOCUS to reveal that early-layer attention patterns serve as a predictor for token decodability in DLLMs, achieving up to 2.32× speed up and 3.52× on large blocks.

We hope to inspire future research with flexible-order parallel decoding.